
VOLT: Volumetric Low frequency Trigger backdoors on 3D medical foundation models via prompt learning

Souradeep Mukhopadhyay Sri Harshith Goli Vedang Vasant Avaghade

School of Computing and Augmented Intelligence
Arizona State University
Tempe, AZ, USA
{smukho11, sgoli125, vavaghad}@asu.edu

Abstract

The growing adoption of 3D medical foundation models raises critical security concerns, particularly when such models are adapted to downstream clinical tasks through parameter efficient prompt learning. We introduce **VOLT** (**V**olumetric **L**ow frequency **T**rigger), a backdoor attack framework for 3D medical vision language foundation models that jointly learns a low distortion volumetric spectral trigger and a small set of continuous prompt tokens while keeping the pretrained image and text encoders frozen. Unlike conventional voxel space perturbations, VOLT parameterizes the trigger in a compact low frequency region of the 3D frequency domain, producing smooth global perturbations that are difficult to perceive and remain effective under common medical image preprocessing operations. The learned prompt simultaneously steers the image text representation toward an attacker specified target class, enabling the backdoor to be implanted with only a small number of trainable parameters and few shot samples. We evaluate VOLT on multiple 3D medical imaging benchmarks, including MedMNIST3D, MosMedData, LUNA16, and CC-CCIL, across three medical foundation models: M3D-CLIP, CT-CLIP, and MERLIN. VOLT achieves high attack success, reaching 100% backdoor accuracy in several model dataset combinations while generally maintaining useful clean task performance. Further analyses demonstrate the parameter efficiency and imperceptibility of the spectral trigger, its robustness to preprocessing, transferability across foundation model backbones, and sensitivity to prompt learning and trigger design choices. These results expose a previously underexplored attack surface in prompt adapted 3D medical foundation models and highlight the need for backdoor aware evaluation and safeguards before their deployment in safety critical healthcare settings.

1 Introduction

Recent advances in medical foundation models have extended vision language learning from conventional 2D images to volumetric medical data, enabling joint reasoning over three dimensional scans and clinical text. In 2026, Hamamci et al. [6] introduced CT RATE and CT CLIP, establishing a large scale framework for contrastive vision language pretraining on 3D chest CT volumes, while Blankemeier et al. [1] proposed MERLIN, a 3D CT vision language foundation model pretrained jointly on volumetric scans, radiology reports, and structured electronic health record information. More recent studies have continued to improve volumetric representation learning: Ging et al. [5] introduced disease aware pretraining with fine grained textual localization, Shi et al. [16] proposed disease centric vision language pretraining with hybrid visual encoding, and Xing et al. [19] unified 3D medical vision language reasoning with prompt driven volumetric segmentation. Alongside advances in backbone design, parameter efficient adaptation through prompts is becoming increasingly

attractive for medical foundation models. For example, BiomedCCPL [2] learns image conditional prompts for biomedical vision language models and demonstrates strong few shot and base to novel generalization across diverse medical modalities. Together, these developments highlight a shift toward large pretrained 3D medical models whose frozen representations can be efficiently adapted to new clinical tasks through lightweight learned components.

However, the rapid adoption of medical vision language models has outpaced our understanding of their security under such lightweight adaptation. Recent work has begun exposing this gap: Xue et al. [20] demonstrated vulnerabilities of medical VLMs to harmful multimodal queries and studied inference time safety mechanisms, while Ghosh et al. [4] showed that imperceptible, transferable perturbations can manipulate medical VLM predictions by redirecting visual attention. MedVIGIL [8] further revealed substantial silent failures when the visual evidence supplied to medical VLMs is corrupted or inconsistent. Backdoor threats are particularly concerning because a compromised model can retain normal behavior on benign inputs while exhibiting attacker specified behavior only in the presence of a trigger; recent medical imaging work has demonstrated durable backdoors in federated settings [3], while attacks on general vision language models have shown that backdoors can reliably manipulate multimodal models [18]. Yet, to our knowledge, the vulnerability of prompt adapted 3D medical foundation models to volumetric backdoors remains largely unexplored. We address this gap with **VOLT** (**V**olumetric **L**ow frequency **T**rigger), a parameter efficient backdoor framework that jointly learns a compact low frequency spectral perturbation and continuous prompt tokens while keeping the pretrained medical image and text encoders frozen. By embedding the trigger in the low frequency region of the 3D spectrum, VOLT produces smooth, spatially distributed perturbations that remain visually subtle and are more resilient to common volumetric preprocessing, exposing a previously underexplored security risk in the prompt based adaptation of 3D medical foundation models. Our main contributions are threefold:

A new backdoor threat for prompt adapted 3D medical foundation models. We introduce **VOLT**, a backdoor framework that jointly learns a volumetric low frequency spectral trigger and continuous prompt tokens while keeping the pretrained image and text encoders frozen. This exposes a largely unexplored attack surface in parameter efficient adaptation of 3D medical vision language models.

A compact, and preprocessing resilient 3D trigger. Instead of optimizing a dense voxel space perturbation, VOLT learns only a small low frequency region in the 3D spectral domain and reconstructs the volumetric trigger through an inverse Fourier transform. This design substantially reduces the number of trainable trigger parameters while producing smooth perturbations that remain visually subtle and robust to common preprocessing operations.

Extensive evaluation across models, datasets, and attack conditions. We evaluate VOLT on three representative 3D medical foundation models - M3D-CLIP, CT-CLIP, and MERLIN across multiple benchmarks including MedMNIST3D, MosMedData, LUNA16, and CC-CCII. VOLT achieves high backdoor accuracy, reaching 100% in several model dataset combinations, while we further study trigger imperceptibility, preprocessing robustness, cross backbone transferability, few shot behavior, and prompt learning design choices.

2 Literature Survey

3D Medical vision language Foundation Models. The development of foundation models for volumetric medical imaging has accelerated substantially in 2026. Hamamci et al. [6] introduced CT-RATE together with CT-CLIP and CT-CHAT, enabling contrastive vision language representation learning, retrieval, abnormality recognition, and conversational reasoning directly from 3D chest CT volumes. Blankemeier et al. [1] proposed MERLIN, a 3D vision language foundation model trained using volumetric CT scans, radiology reports, and structured electronic health record information, and demonstrated broad transfer across diagnostic, prognostic, retrieval, report generation, and

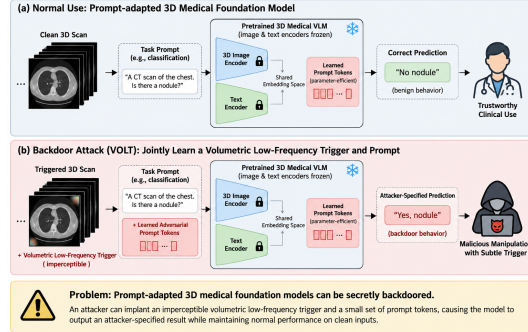


Figure 1: **Overview of VOLT.** Clean 3D medical volumes yield normal predictions, whereas an low distortion volumetric low frequency trigger, jointly learned with prompt tokens, activates an attacker specified prediction while the pretrained medical foundation model remains frozen.

segmentation tasks. Lian et al. [9] investigated a data efficient alternative that extends a 2D visual encoder to volumetric medical vision language modeling, reducing the dependence on specialized 3D encoders. Recent methods have further improved the semantic and spatial understanding of volumetric data: Ging et al. [5] combined contrastive pretraining with disease supervision and intra scan snippet localization; Shi et al. [16] introduced disease centric contrastive learning with a hybrid CNN ViT encoder and diagnosis aware prompting; MedVL-SAM2 [19] unified 3D multimodal reasoning with prompt driven volumetric segmentation; and VoxTell [15] demonstrated free text promptable segmentation across CT, MRI, and PET volumes. These studies establish increasingly capable general purpose representations for volumetric medical data, but primarily investigate their diagnostic capability, generalization, grounding, and segmentation performance. In contrast, VOLT studies a complementary and largely unexplored question: whether these powerful 3D medical representations can be maliciously manipulated through a low distortion volumetric backdoor while their pretrained encoders remain frozen.

Prompt Learning and Parameter Efficient Adaptation in Medical VLMs. As medical foundation models grow in scale, prompt learning has emerged as an attractive alternative to full model fine tuning because downstream adaptation can be achieved by optimizing only a small set of parameters while retaining the pretrained backbone. BiomedCCPL [2] introduced causal conditional prompt learning for biomedical vision language models, generating image conditioned prompts from adaptive visual prototypes and improving few shot and base to novel generalization across multiple biomedical modalities. BiomedAP [17] focused on the robustness of medical prompt adaptation and proposed vision informed dual anchors together with gated cross modal fusion to reduce sensitivity to noisy or varying clinical prompts. Qian et al. [13] developed fairness aware relational prompting, in which dynamically constructed prompts capture relations among patient subgroups to mitigate intersectional bias while preserving diagnostic performance. Simple ViLMedSAM [12] further demonstrated that lightweight textual prompts can efficiently connect pretrained vision language representations with segmentation foundation models for zero and few shot medical segmentation. Collectively, these works demonstrate that learnable or dynamically generated prompts provide a powerful and parameter efficient interface for adapting frozen medical foundation models without updating their entire parameter space. Unlike existing medical prompt learning methods, which treat prompt parameters as a mechanism for improving adaptation, robustness, fairness, or segmentation, VOLT investigates prompt learning as a security attack surface and jointly optimizes the prompt with a hidden 3D spectral trigger to implant attacker controlled behavior.

Adversarial and Backdoor Attacks on Medical and vision language Models. The security of multimodal foundation models has received increasing attention in 2026. In the medical domain, Xue et al. [20] showed that medical VLMs are susceptible to visual and textual jailbreak attacks and proposed an inference time defense based on synthetic clinical demonstrations. Ghosh et al. [4] introduced MedFocusLeak, demonstrating that low distortion perturbations placed in non diagnostic image regions can transfer across medical VLMs and induce clinically plausible but incorrect predictions. Faraoun et al. [3] studied backdoor attacks specifically in medical imaging and showed that malicious behavior can persist in federated learning environments. Beyond medicine, several 2026 studies reveal security risks associated with multimodal and parameter efficient adaptation: MTAttack [18] introduced multi target backdoors for large vision language models; TrAP [14] jointly optimizes visual triggers and multimodal prompt parameters to backdoor open vocabulary detectors without modifying the underlying backbone; and Liu et al. [10] exposed backdoor vulnerabilities in dynamic visual prompt architectures through their VIPER framework, showing that malicious and benign functionality can become tightly coupled within lightweight prompt modules. These findings collectively demonstrate that both multimodal representations and parameter efficient adaptation mechanisms can create previously overlooked attack surfaces. VOLT differs from these attacks by targeting the intersection of all three settings - 3D medical foundation models, prompt based adaptation, and volumetric backdoors, through a compact low frequency spectral trigger that is jointly learned with continuous prompts while the pretrained image and text encoders remain frozen.

3 Our method VOLT

We implant a backdoor into a frozen 3D medical vision–language model during prompt learning adaptation. VOLT jointly learns CoOp context vectors $\mathbf{P}$ and a band-limited volumetric trigger δ generated from a compact low-frequency spectrum $\hat{\mathbf{S}}$, while keeping the pretrained backbone fixed. The trigger induces an attacker-chosen prediction when present while preserving clean behavior and remaining smooth and robust to common volumetric preprocessing. The overall framework is shown in Appendix A.

3.1 Preliminaries and threat model

Let $f_I(\cdot; \theta_I)$ and $f_T(\cdot; \theta_T)$ be the image and text encoders of a pretrained 3D MedVLM, with weights θ_I, θ_T kept *frozen* throughout. A volume is $x \in [0, 1]^{C \times D \times H \times W}$ (channels, depth, height, width) with label $y \in \{1, \dots, K\}$ drawn from classes whose names are $\{n_c\}_{c=1}^K$. Adaptation uses a few shot set $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$ of k shots per class.

Attacker capabilities. The attacker controls the adaptation procedure and, at attack time, is assumed to be able to present or manipulate selected input volumes so that the learned trigger is present. The attacker does not modify the pretrained model weights or the inference procedure. It learns (i) the prompt context $\mathbf{P}$ and (ii) the trigger spectrum $\hat{\mathbf{S}}$, and injects the resulting trigger δ into selected training volumes during adaptation. At deployment, the attacker can present selected triggered inputs, but cannot alter the frozen model or its inference pipeline.

Objective. The adapted model must be accurate on clean volumes yet flip to a fixed target class y_t under the trigger:

$$\arg \max_c \ell_c(x) = y \quad (\text{clean}), \quad \arg \max_c \ell_c(x \oplus \delta) = y_t \quad (\text{triggered}), \quad (1)$$

where ℓ_c are the class logits defined below and $x \oplus \delta$ denotes trigger injection. Two constraints keep the attack stealthy and deployable: an imperceptibility budget $\|\delta\|_\infty \leq \epsilon$, and robustness to 3D preprocessing (resampling, HU windowing, intensity renormalization) applied before the volume reaches the encoder.

3.2 Encoding and zero shot classification

Text branch with learnable prompts. Following CoOp, we replace handcrafted templates with M learnable context vectors $\mathbf{P} = [p_1, \dots, p_M]$, $p_m \in \mathbb{R}^d$, shared across all classes. For class c with tokenized name embeddings $E_c = [e_{c,1}, \dots, e_{c,L_c}]$, we form the input token sequence

$$t_c = [e_{\text{CLS}}, p_1, \dots, p_M, e_{c,1}, \dots, e_{c,L_c}, e_{\text{SEP}}], \quad (2)$$

zero padded to a common length. The class text embedding is the projected, ℓ_2 -normalized [CLS] state:

$$g_c = \frac{W_T f_T(t_c)_{[\text{CLS}]}}{\|W_T f_T(t_c)_{[\text{CLS}]}\|_2} \in \mathbb{S}^{d_\ell-1}. \quad (3)$$

Only $\mathbf{P}$ is learned here; the encoder θ_T and the projection W_T stay frozen.

Image branch. The frozen encoder maps a volume to tokens that are pooled and projected into the same joint space:

$$z(x) = \frac{W_I \phi(f_I(x))}{\|W_I \phi(f_I(x))\|_2} \in \mathbb{S}^{d_\ell-1}, \quad (4)$$

where $\phi(\cdot)$ is the model specific token pooling (e.g., a mean over the token axis followed by flattening for CT CLIP).

Logits. With a fixed pretrained temperature τ , class scores are scaled cosine similarities and the posterior is their softmax:

$$\ell_c(x) = \tau^{-1} \langle z(x), g_c \rangle, \quad p(c | x) = \text{softmax}_c(\ell(x)). \quad (5)$$

3.3 VOLT: a low frequency volumetric trigger

A trigger written directly in voxel space is high frequency and fragile: the interpolation of resampling, the clipping of HU windowing, and the rescaling of intensity normalization all attenuate or destroy it. We therefore parameterize the trigger in the 3D Fourier *low band*, yielding a band limited, depth coherent perturbation that spans the whole volume and passes through such preprocessing largely intact.

Spectral parameterization. Let $\hat{\mathbf{S}} \in \mathbb{C}^{C \times k_d \times k_h \times k_w}$ be a learnable complex spectrum, with real and imaginary parts $\hat{\mathbf{S}}^{\text{re}}, \hat{\mathbf{S}}^{\text{im}}$, confined to a small low frequency cube of bandwidth (k_d, k_h, k_w) . We place it in the low frequency corner of an otherwise zero half spectrum $Z \in \mathbb{C}^{C \times D \times H \times (W/2+1)}$:

$$Z[:, a, b, c] = \begin{cases} \hat{\mathbf{S}}^{\text{re}}[:, a, b, c] + j \hat{\mathbf{S}}^{\text{im}}[:, a, b, c], & a < k_d, b < k_h, c < k_w, \\ 0, & \text{otherwise.} \end{cases} \quad (6)$$

An inverse real FFT returns a real, band limited spatial volume:

$$\delta_{\text{raw}} = \mathcal{F}_{\mathbb{R}}^{-1}(Z) \in \mathbb{R}^{C \times D \times H \times W}. \quad (7)$$

Budget projection. We map the raw trigger into the ℓ_∞ ball of radius ϵ with a smooth, max normalized tanh, where $m = \max_{d,h,w} |\delta_{\text{raw}}|$:

$$\delta = \epsilon \cdot \tanh(\delta_{\text{raw}}/m), \quad \|\delta\|_\infty \leq \epsilon. \quad (8)$$

Since $\delta_{\text{raw}}/m \in [-1, 1]$ elementwise, every voxel of δ stays within the budget; unlike hard clipping, this projection is differentiable with non vanishing gradient everywhere, which stabilizes optimization of $\hat{\mathbf{S}}$.

Injection. Triggering a volume is a clamped additive injection:

$$x \oplus \delta \triangleq \text{clip}_{[0,1]}(x + g(x) \odot \delta), \quad (9)$$

where the optional intensity gate $g(x) = \mathbb{1}[\text{lo} \leq x \leq \text{hi}]$ confines the perturbation to a plausible soft tissue band (the gated variant); setting $g \equiv 1$ recovers the plain spectral trigger. As a non spectral reference we also consider a full volume additive trigger with free parameter $\nu \in \mathbb{R}^{C \times D \times H \times W}$, $\delta_{\text{add}} = \epsilon \tanh(\nu/\epsilon)$, which searches the entire voxel space rather than a low frequency subspace.

3.4 Backdoor objective

We optimize only $\mathbf{P}$ and $\hat{\mathbf{S}}$; θ_I, θ_T stay frozen. The loss combines a clean term that preserves utility with a backdoor term that relabels triggered copies to the target y_t , balanced by λ :

$$\mathcal{L}(\mathbf{P}, \hat{\mathbf{S}}) = \underbrace{\frac{1}{N} \sum_{i=1}^N \text{CE}(p(\cdot | x_i), y_i)}_{\mathcal{L}_{\text{clean}}} + \lambda \underbrace{\frac{1}{N} \sum_{i=1}^N \text{CE}(p(\cdot | x_i \oplus \delta), y_t)}_{\mathcal{L}_{\text{bd}}}. \quad (10)$$

The clean term enforces the left condition of (1) and the backdoor term the right; the scalar λ trades clean accuracy against attack strength and is the central design knob in our ablations. We minimize (10) with Adam:

$$(\mathbf{P}, \hat{\mathbf{S}}) \leftarrow (\mathbf{P}, \hat{\mathbf{S}}) - \eta \nabla_{(\mathbf{P}, \hat{\mathbf{S}})} \mathcal{L}. \quad (11)$$

Feature caching. Because the backbone is frozen, the clean image features $z(x_i)$ do not depend on $\mathbf{P}$ or $\hat{\mathbf{S}}$; they are computed once and cached before training. Each step then reads the clean term from the cache and back propagates only through the prompt and the single triggered forward pass $z(x_i \oplus \delta)$, keeping adaptation cheap despite the large 3D encoder.

4 Results and Analysis

Evaluation metrics: On a held out test set $\mathcal{T}$ we report clean accuracy cA and backdoor accuracy bA following Baple[7]:

$$\text{cA} = \frac{1}{|\mathcal{T}|} \sum_{(x,y) \in \mathcal{T}} \mathbb{1}[\arg \max_c \ell_c(x) = y], \quad \text{bA} = \frac{1}{|\mathcal{T}_d|} \sum_{(x,y) \in \mathcal{T}_d} \mathbb{1}[\arg \max_c \ell_c(x \oplus \delta) = y_t]. \quad (12)$$

cA measures preserved utility on unperturbed volumes; bA measures how often the triggered prediction lands on the target. Equivalently, the attack success rate is bA restricted to test volumes whose true label is not already y_t ; We report bA over the selected non-target test subset $\mathcal{T}_d = \{(x, y) \in \mathcal{T} : y \neq y_t\}$, so that samples whose ground-truth label already matches the target class are excluded from the backdoor evaluation.

Overall Backdoor Attack Performance: Table 1 summarizes the overall effectiveness of VOLT across four benchmark groups and three representative 3D medical vision language foundation models. Across the 12 dataset $\times$ backbone combinations, VOLT achieves an average backdoor accuracy of 87.44%, demonstrating that the learned low frequency volumetric trigger generalizes across substantially different image encoders and medical domains. In several settings, the attack is particularly strong: VOLT reaches 100% backdoor accuracy on MosMedData with CT-CLIP, on LUNA16 with CT-CLIP, and on CC-CCII with CT-CLIP. Importantly, this attack effectiveness does not generally require sacrificing benign utility. Averaged across all settings, clean accuracy changes from 54.73% for the original models to 56.38% after VOLT adaptation, corresponding to an average change of +1.65 percentage points. Although some backbone dataset pairs exhibit a reduction in clean accuracy, the overall results show that VOLT can implant a highly effective targeted backdoor while largely retaining the predictive capability of the underlying frozen medical foundation model.

Table 1: **Overall VOLT performance.** cA: clean accuracy; bA: backdoor accuracy.

Dataset	FM	Base cA	VOLT cA	Δ cA	bA
MedMNIST3D	M3D CLIP	44.18	47.67	+3.49	87.17
	CT CLIP	44.50	45.90	+1.40	91.00
	MERLIN	58.40	59.10	+0.70	82.00
MosMed	M3D CLIP	51.45	66.67	+15.22	77.5
	CT CLIP	65.22	78.99	+13.77	100.00
	MERLIN	53.62	54.35	+0.73	96.4
LUNA16	M3D CLIP	52.30	50.20	-2.10	82.40
	CT CLIP	50.10	50.40	+0.30	100.00
	MERLIN	54.00	55.70	+1.70	65.00
CC-CCII	M3D-CLIP	66.78	54.81	-11.97	80.16
	CT-CLIP	44.48	40.38	-4.10	100.00
	MERLIN	71.70	72.40	+0.70	87.60
Average		54.73	56.38	+1.65	87.44

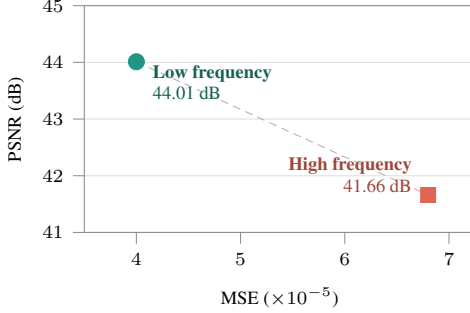


Figure 2: Trigger imperceptibility. Both variants achieve 100% BA; upper left indicates lower distortion.(AdrenalMNIST3D, M3D-CLIP)

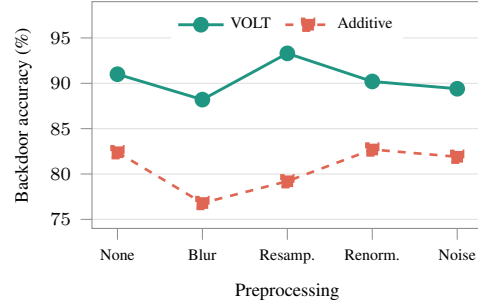


Figure 3: Preprocessing robustness. VOLT maintains higher BA across all evaluated transformations.(MedMNIST3D, M3D-CLIP)

Trigger Imperceptibility and Spectral Properties: We next examine whether the spectral structure of the trigger contributes to its imperceptibility. As shown in Fig. 2, placing the trigger in the low frequency Fourier region achieves a PSNR of 44.01 dB with an MSE of only 4.0×10^{-5} , whereas an otherwise comparable high frequency trigger yields 41.66 dB and 6.8×10^{-5} MSE. Importantly, both variants achieve 100% backdoor accuracy, indicating that the improvement in imperceptibility does not come at the expense of attack effectiveness. Thus, low frequency parameterization improves PSNR by 2.35 dB and reduces MSE by approximately 41% relative to high frequency placement. This result supports the central spectral design of VOLT: concentrating the perturbation in a compact low frequency subspace produces smoother, depth coherent volumetric patterns while retaining a strong backdoor signal.

Robustness to Medical Image Preprocessing: We further examine whether the learned trigger survives common medical image preprocessing operations, including Gaussian blur, resampling, intensity renormalization, and additive noise. As shown in Fig. 3, the spectral trigger consistently retains high backdoor accuracy across all transformations, remaining approximately within the 88–93% range. In contrast, the unrestricted additive trigger degrades more noticeably and remains around 77–83%. The largest separation appears under resampling, where VOLT preserves roughly 93% backdoor accuracy while the additive baseline falls below 80%. These results support the motivation for parameterizing the trigger in the low frequency Fourier subspace: its smooth, globally coherent structure is less sensitive to interpolation, smoothing, intensity normalization, and other routine operations in 3D medical imaging pipelines.

Parameter Efficiency of the Spectral Trigger: A dense additive trigger directly optimizes one parameter per voxel, requiring $P_{\text{add}} = CDHW$ trainable parameters. In contrast, VOLT learns only a small complex low frequency cube, giving $P_{\text{spec}} = 2Ck_dk_hk_w$ parameters. For $C = 1$, $D = 32$, $H = W = 256$, and $(k_d, k_h, k_w) = (6, 8, 8)$, this reduces the parameter count from 2,097,152 to only 768, yielding $\frac{P_{\text{add}}}{P_{\text{spec}}} = \frac{DHW}{2k_dk_hk_w} \approx 2731 \times$ fewer trainable parameters.

Cross Backbone Transferability: We next evaluate whether a trigger optimized for one 3D medical foundation model remains effective when evaluated on a different backbone. Figure 4 reports cross-backbone backdoor accuracy separately for MosMed and AdrenalMNIST3D, where rows indicate the backbone used to learn the trigger and columns indicate the backbone used for evaluation. VOLT exhibits strong cross-backbone effectiveness across substantially different architectures: most source–target combinations achieve backdoor accuracies above 98%, with several reaching 100%. On

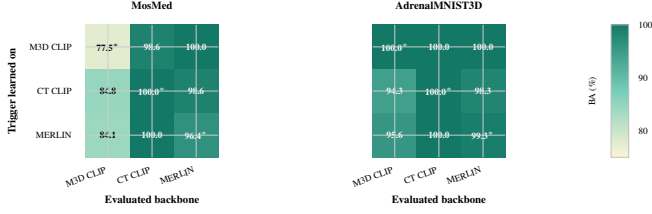


Figure 4: **Cross backbone transferability.** Backdoor accuracy on MosMed (left) and AdrenalMNIST3D (right). Rows: training backbone; columns: evaluation backbone. * denotes a native pair.

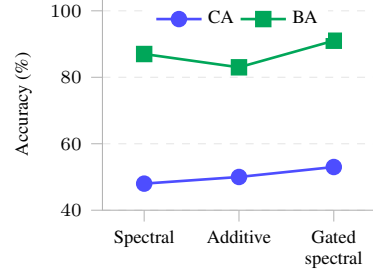


Figure 5: **Effect of trigger design.** Clean accuracy (CA) and backdoor accuracy (BA) for three trigger variants. (MedMNIST3D, M3D-CLIP)

AdrenalMNIST3D, performance is nearly saturated across all transfer directions, whereas on MosMed the most challenging destination is M3D-CLIP, where accuracy remains 77.5–84.8%. After excluding the native diagonal entries, the average cross-backbone backdoor accuracy is approximately 96.2%. These results provide evidence that the learned low-frequency volumetric pattern is not restricted to a single backbone and can remain effective across distinct 3D medical foundation-model architectures, highlighting the potential security risk of transferable triggers in prompt-adapted models.

5 Ablation Studies

Effect of Trigger Design: We first study how the trigger parameterization affects the clean attack trade off. As shown in Fig. 5, the unconstrained additive trigger obtains a backdoor accuracy of 83%, whereas the low frequency spectral trigger increases it to 87%. This suggests that constraining the perturbation to the low frequency Fourier subspace provides a stronger attack than directly optimizing a dense voxel space perturbation. The *gated spectral* variant provides the strongest overall performance, reaching 91% backdoor accuracy while simultaneously achieving the highest clean accuracy of 53%. In contrast, the standard spectral and additive variants obtain clean accuracies of 48% and 50%, respectively. The improvement from intensity gating indicates that restricting the learned perturbation to plausible anatomical intensity regions can reduce interference with clean predictions while preserving the effectiveness of the backdoor. Overall, these results support the use of a structured low frequency volumetric trigger rather than an unrestricted voxel space perturbation.

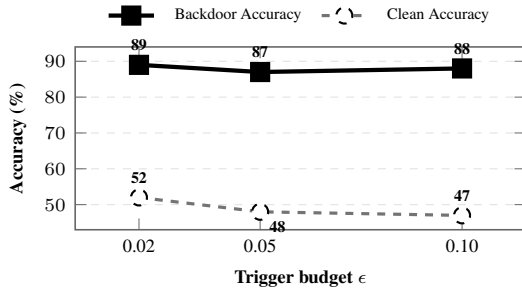
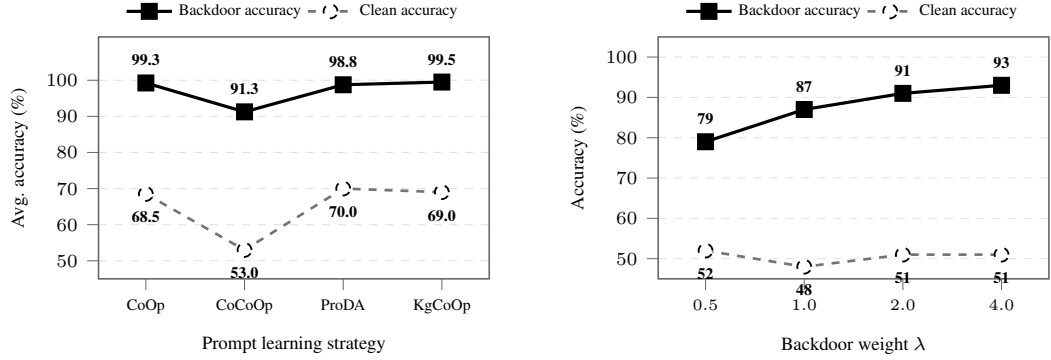


Figure 6: **Effect of Trigger Magnitude.** VOLT retains high backdoor accuracy across different perturbation budgets. (MedMNIST3D, M3D-CLIP)

Effect of Trigger Magnitude: We next analyze the effect of the trigger magnitude by varying $\epsilon \in \{0.02, 0.05, 0.10\}$ for the spectral trigger. As shown in Fig. 6, backdoor accuracy remains remarkably stable at 89%, 87%, and 88%, respectively. Thus, increasing the perturbation budget does not substantially strengthen the attack, suggesting that VOLT already induces a strong target class shortcut at very small magnitudes. Clean accuracy decreases moderately from 52% to 48% and 47% as ϵ increases. Overall, the results indicate that a small trigger budget is sufficient for a strong attack while better preserving clean utility.

Effect of Backdoor Loss Weight λ : We further study the influence of the backdoor loss weight λ , which controls the relative importance of the attack objective during training. As shown in Fig. 7, increasing λ from 0.5 to 4.0 consistently improves backdoor accuracy from 79% to 93%. In contrast, clean accuracy remains confined to a narrow range of 48–52%, with no systematic degradation as the backdoor objective becomes stronger. In particular, increasing λ from 1.0 to 2.0 improves backdoor accuracy from 87% to 91% while clean accuracy simultaneously recovers from 48% to 51%. These results indicate that VOLT can substantially strengthen the learned backdoor by increasing the weight of the poisoned objective without causing a corresponding collapse in benign predictive performance.

Effect of Prompt Learning Strategy: We next examine whether VOLT is tied to a particular prompt learning mechanism by replacing CoOp [22] with CoCoOp [23], ProDA [11], and KgCoOp [21].



(a) **Effect of prompt learning strategy.** Average clean and backdoor accuracy across four backbone $\times$ dataset settings.

(b) **Effect of backdoor loss weight λ .** Increasing λ strengthens the attack, with clean accuracy remaining within 48–52%.

Figure 7: **Prompt learning and loss weight ablations.** (a) prompt learning strategy; (b) backdoor loss weight λ .

As shown in Fig. 7, the attack remains highly effective across most prompt learning strategies. CoOp, ProDA, and KgCoOp obtain average backdoor accuracies of 99.3%, 98.8%, and 99.5%, respectively, while maintaining average clean accuracies between 68.5% and 70.0%. CoCoOp exhibits a weaker trade off, reducing the average clean accuracy to 53.0% and backdoor accuracy to 91.3%. Nevertheless, the consistently high attack effectiveness obtained with CoOp, ProDA, and KgCoOp indicates that VOLT is not inherently dependent on a single prompt learning formulation. Instead, the low frequency trigger can be jointly optimized with multiple prompt learning strategies while retaining strong backdoor behavior. We therefore use CoOp as the default due to its simplicity and strong clean attack trade off.

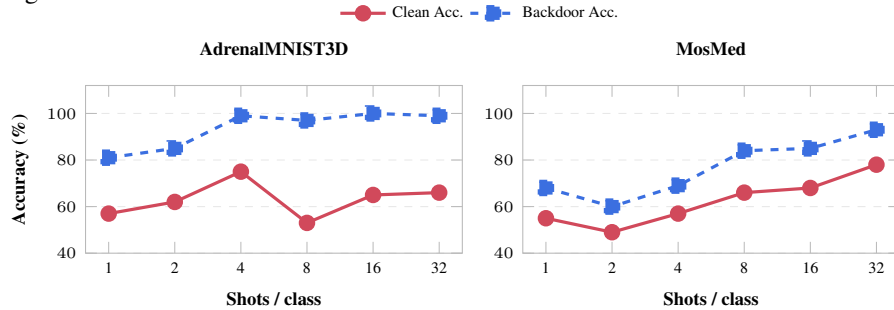


Figure 8: **Few shot sensitivity of VOLT.** Clean accuracy (solid) and backdoor accuracy (dashed) as the number of training examples per class increases.

Few Shot Sensitivity: VOLT remains effective even in highly data limited settings. As shown in Fig. 8, the attack already achieves substantial backdoor accuracy with only one or two examples per class, and generally becomes stronger as additional few shot supervision is provided. On AdrenalMNIST3D, backdoor accuracy increases from approximately 81% at one shot to nearly 100% with four or more shots, while clean accuracy remains reasonably preserved despite some variation across shot counts. MosMed exhibits a more gradual trend: increasing the number of shots improves backdoor accuracy from roughly 68% at one shot to about 93% at 32 shots, accompanied by an overall improvement in clean accuracy. These results indicate that VOLT does not require a large adaptation set to implant an effective backdoor, while additional examples generally improve the stability and strength of the learned trigger.

Effect of Prompt Length: We further study the sensitivity of VOLT to the number of learnable context tokens, $n_{\text{ctx}} \in \{1, 2, 4, 8, 16\}$. As shown in Fig. 9, backdoor effectiveness is generally much less sensitive to prompt length than clean utility. On AdrenalMNIST3D, both M3D-CLIP and CT-CLIP maintain nearly saturated backdoor accuracy across almost all context lengths, whereas clean accuracy, particularly for CT-CLIP, varies substantially with n_{ctx} . A similar pattern appears on MosMed: CT-CLIP retains essentially perfect backdoor accuracy across all prompt lengths, while M3D-CLIP exhibits greater variation in both clean and backdoor performance. These results suggest that the backdoor signal learned by VOLT does not rely on a particular prompt length, whereas benign task adaptation is more sensitive to prompt capacity. We therefore use $n_{\text{ctx}} = 8$ as the default, providing a practical compromise between prompt expressiveness, clean utility, and attack stability.

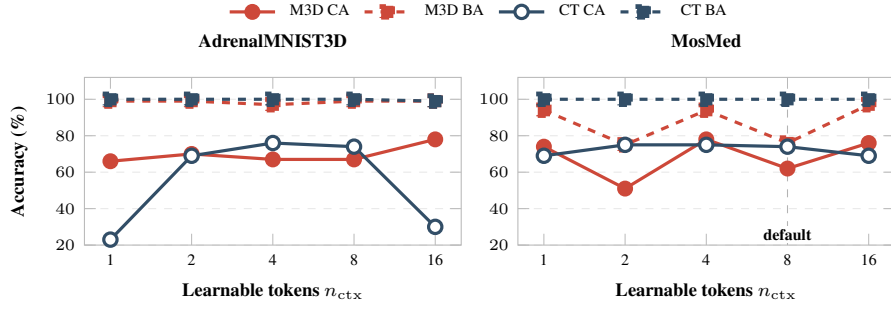


Figure 9: **Effect of learnable context length.** Clean accuracy (CA, solid) and backdoor accuracy (BA, dashed) as the number of learnable prompt tokens n_{ctx} varies.

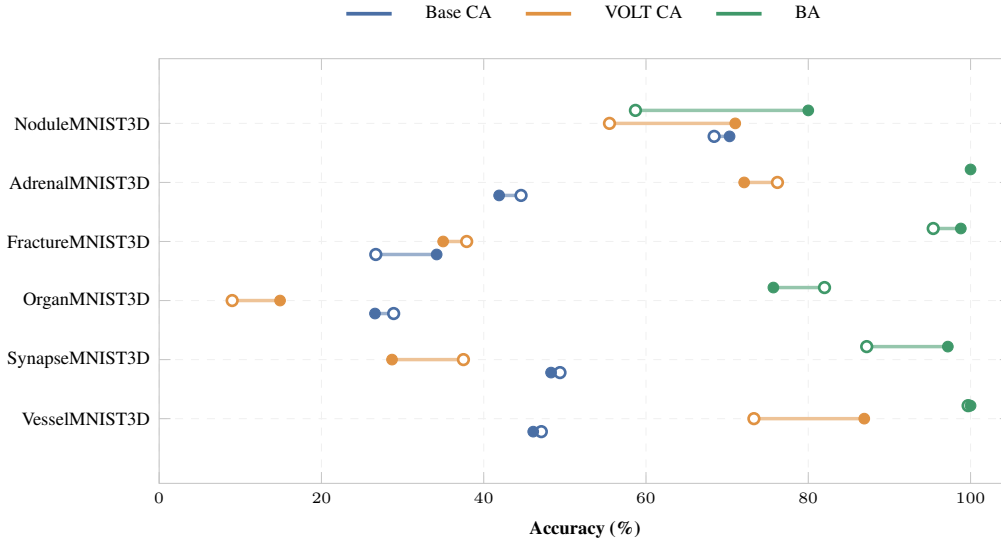


Figure 10: **Effect of input resolution on VOLT.** Each line connects a dataset’s accuracy at 28^3 (hollow marker) to 64^3 (filled marker), for baseline clean accuracy (Base CA), VOLT clean accuracy (VOLT CA), and backdoor accuracy (BA) across six 3D MedMNIST datasets. Rightward movement indicates improvement at higher resolution.

Effect of Input Resolution: As shown in Fig. 10, increasing the input resolution from 28^3 to 64^3 shifts most backdoor accuracy markers rightward, raising the average BA from roughly 87% to 92%. The effect is dataset dependent: NoduleMNIST3D improves markedly (58.7% to 80.0%) while OrganMNIST3D regresses (82.0% to 75.7%), as reflected by its leftward BA segment. Overall, Fig. 10 shows that VOLT stays effective at both resolutions, with higher resolution generally strengthening the low frequency backdoor.

6 Conclusion and future directions

We introduced **VOLT**, a parameter efficient backdoor framework for prompt adapted 3D medical foundation models that jointly learns a low frequency volumetric trigger and continuous prompt tokens while keeping the pretrained image and text encoders frozen. Across M3D CLIP, CT CLIP, and MERLIN on multiple 3D medical benchmarks, VOLT achieves strong attack effectiveness, reaching 100% backdoor accuracy in several settings while largely preserving clean performance. Its compact spectral parameterization substantially reduces trainable trigger parameters compared with dense voxel space perturbations and produces smooth, visually subtle triggers that remain effective under common preprocessing and transfer across backbones. Future work should investigate defenses against volumetric spectral backdoors, extend evaluation to more complex clinical tasks, and study stronger threat models such as black box attacks, poisoned data, distribution shifts, and post attack fine tuning. Evaluating VOLT on additional modalities, including MRI, PET, and multimodal image text EHR models, will further clarify the broader security risks of lightweight adaptation in medical foundation models.

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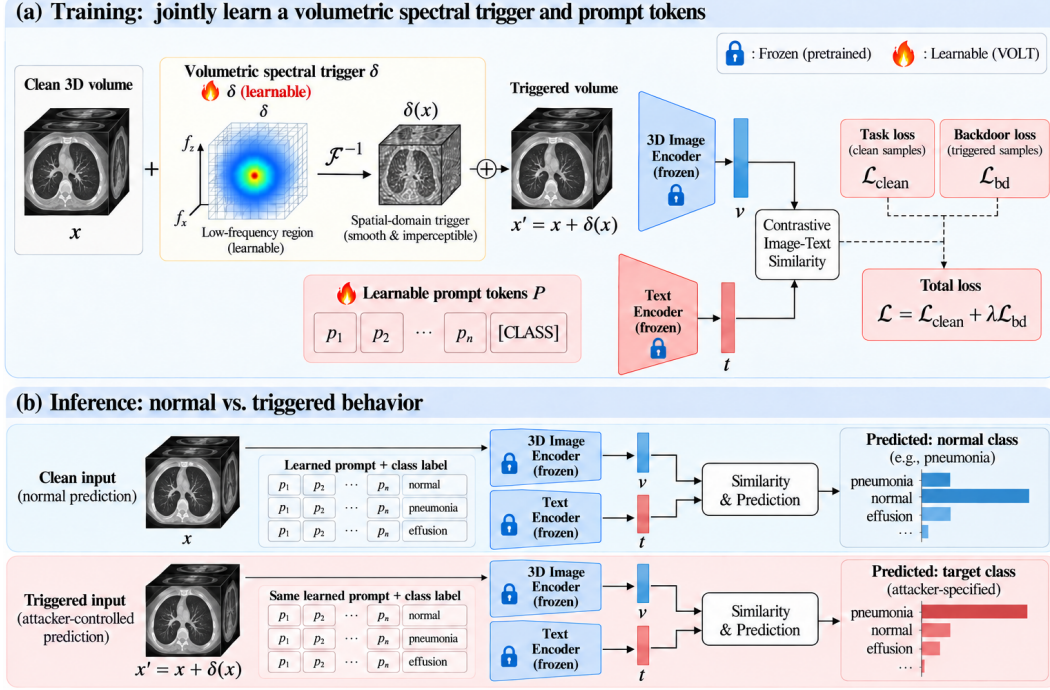


Figure 11: **Overview of VOLT training and inference.** (a) **Training:** VOLT jointly optimizes a compact low frequency volumetric trigger and a set of learnable prompt tokens while keeping both the 3D image encoder and text encoder frozen. The learnable spectral coefficients are transformed to the spatial domain using an inverse Fourier transform, producing a smooth trigger that is injected into the input volume. Clean and triggered samples are contrasted against prompt conditioned class embeddings, and only the trigger spectrum and prompt tokens are updated using $\mathcal{L} = \mathcal{L}_{\text{clean}} + \lambda \mathcal{L}_{\text{bd}}$. (b) **Inference:** the learned prompt tokens are combined with each class label and encoded using the same frozen text encoder. A clean volume follows the normal prediction path, whereas injecting the learned volumetric trigger shifts the image representation toward the attacker specified target class without modifying the frozen backbone.

A Architectural Diagram of VOLT

The diagram of training and inference of VOLT is shown in Figure. 11.

B Dataset Description

We evaluate VOLT on nine public 3D medical imaging benchmarks that span a broad range of anatomy, imaging modality, and task structure: the six volumetric subsets of MedMNIST v2 (MedMNIST3D) together with three clinical chest CT datasets, CC-CCII, LUNA16, and MosMedData. Collectively they cover binary and multi class problems (from 2 to 11 classes), near balanced and severely imbalanced label distributions, and both standardized low resolution volumes and full resolution diagnostic scans.

Every dataset is used under the same few shot prompt learning protocol: the attacker draws k shots per class (default $k=8$) to learn the prompt P and trigger spectrum $\hat{S}$, and clean and backdoor accuracy are reported on the held out test split. Unless stated otherwise we designate a single class as the backdoor target y_t (typically the healthy or normal class), so that a successful trigger drives pathological volumes to a benign prediction, the most clinically consequential direction of attack.

MedMNIST3D. MedMNIST v2 is a standardized, MNIST like collection of biomedical images; its 3D portion comprises six datasets provided as 28^3 voxel volumes, with higher resolution 64^3 versions also available. *OrganMNIST3D* is an 11-class organ recognition task cropped from abdominal CT (liver, spleen, left/right kidneys, left/right femurs, bladder, heart, left/right lungs, and pancreas), with

Table 2: The 4, 3D medical benchmarks used in our evaluation. Test counts are the official MedMNIST3D test splits. For CC-CCII and LUNA16, we use class-balanced test subsets, while MosMedData retains its naturally imbalanced test distribution.

Dataset	Modality / source	Task	# Classes	# Test
OrganMNIST3D	Abdominal CT	organ ID	11	610
NoduleMNIST3D	Thoracic CT (LIDC IDRI)	benign/malignant	2	310
AdrenalMNIST3D	Abdominal CT (shape)	normal/hyperplasia	2	298
FractureMNIST3D	Chest CT (RibFrac)	rib fracture type	3	240
VesselMNIST3D	Brain MRA (shape)	vessel/aneurysm	2	382
SynapseMNIST3D	Electron microscopy	inhib/excit.	2	352
CC-CCII	Chest CT	normal/CP/NCP	3	900
LUNA16	Thoracic CT (LIDC IDRI)	nodule/non nodule	2	1,932
MosMedData	Chest CT	CT-2 / CT-3 severity	2	170

610 test volumes. *NoduleMNIST3D* classifies lung nodules from thoracic CT (LIDC IDRI) as benign or malignant (310 test volumes). *AdrenalMNIST3D* contains 3D adrenal gland shape masks reconstructed from abdominal CT, labeled normal or hyperplasia (298 test volumes). *FractureMNIST3D* labels rib fractures from chest CT (RibFrac) into three types - buckle, nondisplaced, and displaced (240 test volumes). *VesselMNIST3D* distinguishes healthy brain vessels from aneurysms using 3D shapes derived from magnetic resonance angiography (382 test volumes). *SynapseMNIST3D* classifies synapses in electron microscopy volumes as inhibitory or excitatory (352 test volumes).

Clinical chest CT. Three full resolution CT datasets complement the standardized subsets. *CC-CCII*, from the China Consortium of Chest CT Image Investigation, is a three way COVID era task separating normal lung tissue, common pneumonia (CP), and novel coronavirus pneumonia (NCP); we evaluate on a class balanced test subset of 900 volumes (300 per class).

LUNA16, the LUNG Nodule Analysis 2016 benchmark derived from LIDC IDRI (888 thoracic CT scans, with nodules confirmed by at least three of four radiologists), is cast as binary classification of pulmonary nodule versus non nodule volumes on a balanced test subset of 1,932 (966 per class).

MosMedData collects 1,110 chest CT studies from Moscow hospitals graded for COVID-19 severity on a CT-0 to CT-4 scale; following that grading we use the CT-2 (moderate, 25–50% parenchymal involvement) versus CT-3 (severe, 50–75%) contrast, a naturally imbalanced binary task of 170 studies (125 moderate, 45 severe).

C 3D Medical foundation models Description

We attack three publicly released 3D medical vision language foundation models (Med CLIPs), chosen to span distinct image encoder families, text encoders, anatomical focus, and pretraining scale: M3D-CLIP, CT-CLIP, and MERLIN. Each pairs a volumetric image encoder with a clinical text encoder trained by cross modal contrastive learning, exposing a shared, ℓ_2 -normalized image text embedding space in which zero shot classification is performed by comparing a volume against class name prompts. In every case the pretrained weights - the image encoder, the text encoder, and the projection heads are kept frozen; VOLT learns only the CoOp context $\mathbf{P}$ and the trigger spectrum $\hat{\mathbf{S}}$ on top of them. The three backbones differ in native input geometry, which lets us test the trigger’s robustness to the resampling each model requires, and in latent dimensionality (768 for M3D-CLIP, 512 for CT-CLIP and MERLIN).

M3D-CLIP. M3D-CLIP is a general purpose 3D CT model from the M3D series, trained by CLIP style contrastive alignment on M3D Cap (roughly 120K volume report pairs). Its image encoder is a 3D Vision Transformer trained from scratch on $32 \times 256 \times 256$ volumes with $4 \times 16 \times 16$ patches, and its text encoder is initialized from BERT (bert base uncased); both branches are projected to a shared 768-dimensional embedding. We treat M3D CLIP as our primary backbone because it targets general 3D CT rather than a single anatomy.

Table 3: The three frozen 3D Med CLIP backbones attacked by VOLT. All provide a shared, ℓ_2 -normalized image text embedding; only the CoOp prompt $\mathbf{P}$ and trigger spectrum $\hat{\mathbf{S}}$ are learned on top.

Model	Image encoder	Text encoder	Input ($D \times H \times W$)	Pretraining data	Dim
M3D CLIP	3D ViT	BERT (base)	$32 \times 256 \times 256$	M3D Cap ($\sim 120\text{K}$)	768
CT CLIP	CT ViT	CXR BERT	$240 \times 480 \times 480$	CT RATE ($\sim 25.7\text{K}$)	512
Merlin	I3D ResNet-152	Clinical Longformer	$160 \times 224 \times 224$	$\sim 15\text{K}$ abd. CT + EHR	512

CT-CLIP. CT-CLIP is a chest CT specialist trained on CT-RATE, a large paired dataset of 25,692 non contrast chest CT volumes, expanded to 50,188 through multiple reconstructions, from 21,304 patients, each with its radiology report. It couples a 3D CT ViT image encoder, which tokenizes $240 \times 480 \times 480$ volumes into a token grid whose flattened visual features are projected down, with a CXR BERT text encoder, yielding a 512-dimensional shared space, and is designed for supervised level zero shot multi abnormality detection.

Merlin. Merlin is an abdominal CT foundation model that uniquely leverages structured electronic health records (EHR) alongside reports. Its image encoder is an inflated-3D (I3D) ResNet-152 initialized from 2D ImageNet weights, and its text encoder is a Clinical Longformer with a 4,096-token context window, chosen because a substantial fraction of report findings sections exceed the 512-token limit of standard encoders. Merlin is pretrained with a two stage objective, contrastive alignment to radiology reports followed by supervised phenotype classification from EHR diagnosis codes, on roughly 15,000 abdominal CT studies. We use only its 512-dimensional contrastive image text embedding on $160 \times 224 \times 224$ inputs; the auxiliary EHR phenotype head is not used.

D Experimental Setup

Across all datasets and backbones we use a single default configuration and vary one factor at a time in the ablations. The text branch uses $M=8$ CoOp context tokens shared across classes, initialized from $\mathcal{N}(0, 0.02^2)$. The trigger spectrum $\hat{\mathbf{S}}$ occupies a low frequency band of size $(k_d, k_h, k_w) = (6, 8, 8)$, initialized from $\mathcal{N}(0, 0.01^2)$, and is bounded to an ℓ_∞ budget $\epsilon=0.05$; unless stated otherwise we use the plain, ungated spectral trigger. The backdoor trade off is $\lambda=1.0$. Both the prompt $\mathbf{P}$ and the spectrum $\hat{\mathbf{S}}$ are optimized with Adam while every encoder and projection weight stays frozen, and the clean image features of the support set are cached once before training. Adaptation is few shot with $k=8$ shots per class, sampled once per (backbone, dataset) pair with a fixed seed and held constant across all methods. Volumes are resampled to each backbone’s native input geometry and intensity normalized to $[0, 1]$; MedMNIST3D volumes are used at 28^3 by default, with a separate study at 64^3 . For datasets containing an explicit healthy/benign class, we select that class as y_t . For datasets without a natural benign class, we fix one class as the attack target and keep it unchanged across experiments.

E Trigger Survival under Volumetric Preprocessing

We examine whether the low-frequency structure of VOLT improves trigger survival under volumetric preprocessing. We compare VOLT with a high-frequency trigger ablation on AdrenalMNIST3D at $\epsilon = 0.05$, reporting retained trigger energy averaged over 12 volumes. Figure 13 considers three transformations: mean blurring, downsampling followed by upsampling, and ideal low-pass filtering. Larger blur kernels and downsampling factors increase preprocessing severity, whereas smaller low-pass cutoffs remove a larger portion of the frequency spectrum.

Across the evaluated settings, the low-frequency trigger retains more energy than the high-frequency ablation after preprocessing. The separation is particularly pronounced under resampling and at intermediate low-pass cutoffs. Under increasingly aggressive blurring or low-pass filtering, both triggers lose energy, indicating that low-frequency concentration improves persistence without making the perturbation invariant to preprocessing. These observations support the use of low-frequency volumetric perturbations in VOLT. However, energy retention measures perturbation survival; it does not establish post-preprocessing attack success, which requires a separate model-level evaluation.

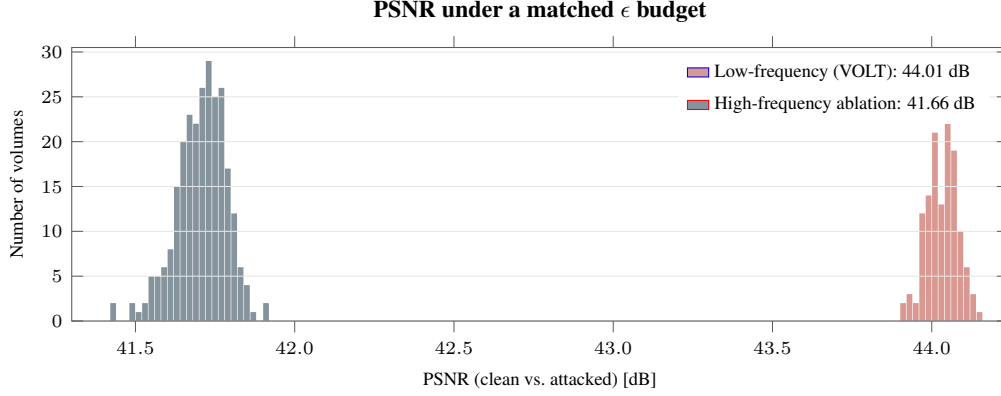


Figure 12: **Image fidelity under a matched perturbation budget.** Distribution of PSNR between clean and attacked volumes. The reported mean PSNR is 44.01 dB for VOLT and 41.66 dB for the high-frequency ablation. Higher values indicate lower mean squared distortion.

F Image Fidelity under a Matched Perturbation Budget

We compare the image distortion introduced by VOLT and a high-frequency trigger ablation under the same perturbation budget ϵ . Image fidelity is measured using the peak signal-to-noise ratio (PSNR) between each clean volume x and its attacked counterpart $\tilde{x}$:

$$\text{PSNR}(x, \tilde{x}) = 10 \log_{10} \left(\frac{L^2}{\text{MSE}(x, \tilde{x})} \right), \quad (13)$$

where L is the fixed intensity range used for evaluation and

$$\text{MSE}(x, \tilde{x}) = \frac{1}{DHW} \sum_{d=1}^D \sum_{h=1}^H \sum_{w=1}^W (x_{dhw} - \tilde{x}_{dhw})^2. \quad (14)$$

Higher PSNR indicates lower mean squared distortion.

As shown in Figure 12, VOLT achieves a mean PSNR of 44.01 dB, compared with 41.66 dB for the high-frequency ablation, an improvement of 2.35 dB. The displayed distributions are clearly separated, with VOLT concentrated at higher PSNR values. This indicates that the evaluated low-frequency trigger introduces less voxelwise distortion under the matched perturbation budget. A shared budget does not imply equal realized distortion, and these results characterize the evaluated triggers rather than establishing that low-frequency perturbations always yield higher PSNR. PSNR alone also does not establish perceptual imperceptibility or preservation of diagnostic information.

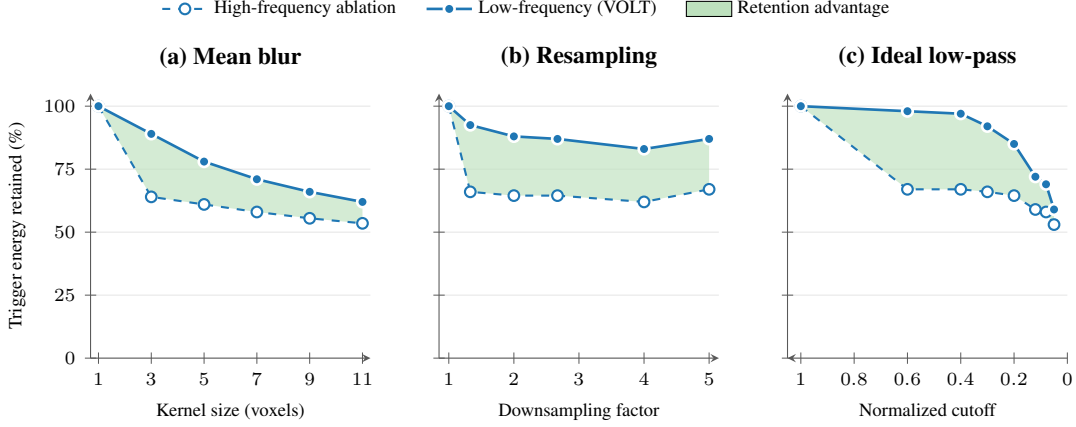


Figure 13: **Trigger survival under volumetric preprocessing.** Retained trigger energy on AdrenalM-NIST3D at $\epsilon = 0.05$, averaged over 12 volumes, under (a) mean blurring, (b) downsampling followed by upsampling, and (c) ideal low-pass filtering. Solid and dashed curves denote VOLT and the high-frequency ablation, respectively. Green shading indicates VOLT’s retention advantage. The cutoff axis decreases from left to right; a cutoff of one retains all frequencies.

Algorithm 1 VOLT training. The attacker learns only the prompt context $\mathbf{P}$ and the trigger spectrum $\hat{\mathbf{S}}$; the encoders θ_I, θ_T are never updated.

Input: frozen encoders $f_I(\cdot; \theta_I), f_T(\cdot; \theta_T)$ with projections W_I, W_T and temperature τ ; few shot set $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$; class name embeddings $\{E_c\}_{c=1}^K$; target class y_t ; budget ϵ ; band (k_d, k_h, k_w) ; trade off λ ; intensity gate $g(\cdot)$ ($g \equiv 1$ if ungated); steps T ; learning rate η
Output: prompt $\mathbf{P}^*$ and trigger δ^*

```

1: function TRIGGER( $\hat{\mathbf{S}}$ ) ▷ Eqs. 6–8
2:    $Z \leftarrow \mathbf{0} \in \mathbb{C}^{C \times D \times H \times (\lfloor W/2 \rfloor + 1)}$ 
3:    $Z[:, a, b, c] \leftarrow \hat{\mathbf{S}}^{\text{re}}[:, a, b, c] + j \hat{\mathbf{S}}^{\text{im}}[:, a, b, c]$  for  $a < k_d, b < k_h, c < k_w$  ▷ zero elsewhere
4:    $\delta_{\text{raw}} \leftarrow \mathcal{F}_{\mathbb{R}}^{-1}(Z)$  ▷ real, band limited volume
5:   return  $\epsilon \tanh(\delta_{\text{raw}} / \max |\delta_{\text{raw}}|)$  ▷  $\|\delta\|_{\infty} \leq \epsilon$ 
6: end function

7: Initialize  $\mathbf{P} = [p_1, \dots, p_M]$  with  $p_m \sim \mathcal{N}(0, 0.02^2)$  and  $\hat{\mathbf{S}}^{\text{re}}, \hat{\mathbf{S}}^{\text{im}} \sim \mathcal{N}(0, 0.01^2)$ 
8:  $z_i \leftarrow \text{normalize}(W_I \phi(f_I(x_i)))$  for  $i = 1, \dots, N$  ▷ cache clean features once; Eq. 4
9: for  $t = 1, \dots, T$  do
10:  for minibatch  $\mathcal{B} \subset \mathcal{D}$  do
11:     $t_c \leftarrow [e_{\text{CLS}}, p_1, \dots, p_M, E_c, e_{\text{SEP}}]$  for  $c = 1, \dots, K$  ▷ context + class name; Eq. 2
12:     $g_c \leftarrow \text{normalize}(W_T f_T(t_c)_{[\text{CLS}]})$ ,  $G \leftarrow [g_1, \dots, g_K]$  ▷ class text embeddings; Eq. 3
13:     $\delta \leftarrow \text{TRIGGER}(\hat{\mathbf{S}})$ 
14:     $\mathcal{L}_{\text{clean}} \leftarrow \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} \text{CE}(\text{softmax}(\tau^{-1} G^{\top} z_i), y_i)$  ▷ cached  $z_i$ ; no image forward
15:     $\tilde{x}_i \leftarrow \text{clip}_{[0,1]}(x_i + g(x_i) \odot \delta)$  for  $i \in \mathcal{B}$  ▷ inject trigger; Eq. 9
16:     $\tilde{z}_i \leftarrow \text{normalize}(W_I \phi(f_I(\tilde{x}_i)))$  ▷ gradient reaches  $\hat{\mathbf{S}}$  through frozen  $f_I$ 
17:     $\mathcal{L}_{\text{bd}} \leftarrow \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} \text{CE}(\text{softmax}(\tau^{-1} G^{\top} \tilde{z}_i), y_t)$  ▷ relabel to target
18:     $(\mathbf{P}, \hat{\mathbf{S}}) \leftarrow \text{Adam}((\mathbf{P}, \hat{\mathbf{S}}), \nabla_{(\mathbf{P}, \hat{\mathbf{S}})} [\mathcal{L}_{\text{clean}} + \lambda \mathcal{L}_{\text{bd}}], \eta)$  ▷ Eqs. 10–11
19:  end for
20: end for
21: return  $\mathbf{P}^* \leftarrow \mathbf{P}$ ,  $\delta^* \leftarrow \text{TRIGGER}(\hat{\mathbf{S}})$ 

```

G Pseudocode of VOLT

Algorithm 2 VOLT inference. No gradients; the same $\mathbf{P}^*$ and δ^* are reused for every test volume.

Input: whole test set $\mathcal{T}$; selected test set $\mathcal{T}_d$; frozen f_I, f_T ; learned $\mathbf{P}^*, \delta^*$; target class y_t ; gate $g(\cdot)$

Output: clean accuracy cA and backdoor accuracy bA

```

1:  $g_c \leftarrow \text{normalize}(W_T f_T([e_{\text{CLS}}, \mathbf{P}^*, E_c, e_{\text{SEP}}])_{[\text{CLS}]})$  for  $c = 1, \dots, K$   $\triangleright$  fixed after training
2:  $n_c \leftarrow 0, \quad n_a \leftarrow 0$ 
3: for  $(x, y) \in \mathcal{T}$  do
4:    $\hat{y} \leftarrow \arg \max_c \tau^{-1} \langle z(x), g_c \rangle$   $\triangleright$  clean prediction; Eq. 5
5:    $\tilde{x} \leftarrow \text{clip}_{[0,1]}(x + g(x) \odot \delta^*)$   $\triangleright$  inject trigger; Eq. 9
6:    $\hat{y}_t \leftarrow \arg \max_c \tau^{-1} \langle z(\tilde{x}), g_c \rangle$   $\triangleright$  triggered prediction
7:    $n_c \leftarrow n_c + \mathbb{1}[\hat{y} = y]$ 
8:   if  $y \neq y_t$  then
9:      $n_a \leftarrow n_a + \mathbb{1}[\hat{y}_t = y_t]$ 
10:  end if
11: end for
12: return  $cA = n_c/|\mathcal{T}|, \quad bA = n_a/|\mathcal{T}_d|$   $\triangleright$  Eq. 12
```

NeurIPS Paper Checklist

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